AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

Student Academic Performance Analysis and Grade Distribution

Project Domain

Excel with Copilot

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

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3. Problem Statement

1. Background of the Chosen Domain and Data-Related Challenges

Educational institutions collect large volumes of student data such as attendance percentage, study hours, exam scores, parental involvement, and motivation levels. However, this data is often unstructured and not systematically analyzed. The selected dataset contains approximately 6500 records and 20 variables, requiring proper cleaning, standardization, and analytical processing to generate meaningful insights.

2. Significance of the Problem in a Real-World Context

Student academic performance directly affects institutional rankings, student progression, and career opportunities. Without data-driven evaluation, institutions rely on assumptions instead of measurable evidence to design academic strategies. This reduces the effectiveness of performance improvement initiatives and resource allocation.

3. Specific Gaps or Inefficiencies Addressed

The key gaps identified are:

• No clear identification of which factors (attendance, study hours, motivation, tutoring, etc.) have the strongest impact on exam scores.
• Absence of a structured grading model to categorize students into A/B/C/D/Fail groups for performance evaluation.
• Lack of an early warning mechanism to detect students with low attendance, weak previous scores, or poor engagement.
• Time-consuming manual reporting methods that increase the risk of calculation errors.
• Limited practical integration of AI tools to generate automated insights and performance summaries.

This project addresses these inefficiencies through structured data analysis and AI-assisted interpretation.

4. Stakeholders and Value Delivered

The solution provides measurable value to:

• **Students** – through early identification of academic risk and timely intervention support.
• **Teachers** – through evidence-based insights on performance drivers to guide mentoring strategies.
• **Administrators** – through accurate grade distribution reports and data-backed decision-making.

The project delivers a clean dataset, structured grade classification, identification of high-impact performance factors, and a data-driven student segmentation model supported by AI-generated analytical summaries.

This enables proactive academic planning rather than reactive problem-solving.

4. Objectives

• To explore and understand the Student Performance Factors dataset containing approximately 6500 records and 20 variables.

• To clean and preprocess the dataset by handling missing values, correcting inconsistent categorical entries, and standardizing data formats.

• To classify students into grade categories (A, B, C, D, Fail) based on exam score ranges and analyze overall grade distribution.

• To identify key performance drivers such as study hours, attendance, sleep hours, motivation level, parental involvement, and tutoring sessions.

• To segment students into high performers, average performers, and at-risk categories for targeted academic intervention.

• To use Excel features such as Pivot Tables, charts, IF functions, COUNTIF, SUMIF, and conditional formatting for analysis.

• To integrate AI tools like Microsoft Copilot or AI Chatbot for generating summaries, insights, and automated interpretations of results.

• To develop a structured analytical report and dashboard presenting clear and actionable insights.

5. Methodology / Implementation Plan

Phase	Stage	Description
1	**Data Collection & Exploration**	The Student Performance Factors dataset containing approximately 6500 records and 20 variables will be imported into Excel. The structure, data types (numerical, categorical, binary), and row count will be

		examined. Initial exploration will include checking distribution of exam scores and understanding variable definitions.
2	**Data Cleaning & Preprocessing**	Duplicate records will be removed. Missing or blank values will be handled appropriately. Categorical variables such as Motivation Level, Family Income, and School Type will be standardized to ensure consistency. A new Grade column will be created using IF formulas to classify students into A, B, C, D, and Fail categories. Data validation and formatting corrections will be performed.
3	**Analysis & AI Integration**	Pivot Tables and Excel functions such as COUNTIF, SUMIF, and IF will be applied to analyze grade distribution and identify relationships between factors (attendance, study hours, tutoring, etc.) and exam scores. Students will be segmented into high performers, average performers, and at-risk groups. Microsoft Copilot or AI Chatbot will be used to generate summaries and interpret key findings.
4	**Visualization / Dashboard Development**	Charts and visual representations such as bar graphs and comparison charts will be created to display grade distribution and factor impact. A structured dashboard will be developed showing KPIs including average score, percentage of high performers, and percentage of at-risk students.
5	**Documentation & Submission**	The complete project report will be prepared. All files including cleaned dataset, Excel analysis file, and documentation will be uploaded to GitHub. A video demonstration explaining the methodology and findings will be recorded and submitted.

6. Resources and Timeline

A. Hardware & Software Requirements

- Laptop or PC — Intel Core i3 or higher, minimum 4 GB RAM (8 GB recommended)
- Operating System: Windows 10 / 11 or macOS
- Microsoft Excel (Microsoft 365 / Excel 2019 or later)
- Power BI Desktop — latest version (free download from Microsoft) — where applicable
- Microsoft Copilot — via Microsoft 365 subscription — where applicable
- AI Chatbot (ChatGPT / Google Gemini / Microsoft Copilot Web) — for prompt-based projects
- Internet connection — required for Copilot and AI Chatbot access
- Web browser — Google Chrome or Microsoft Edge (latest version)

B. Dataset

The dataset used for this project is titled **"Student Performance Factors"**, containing approximately **6500 student records** and **20 variables** related to academic performance, behavioral traits, and socio-economic background.

The dataset consists of the following types of variables:

Numerical Variables (Continuous/Integer):
Hours Studied, Attendance, Sleep Hours, Previous Scores, Tutoring Sessions, Physical Activity, Exam Score

Categorical Variables (Text/Ordinal):
Gender, School Type, Family Income, Motivation Level, Teacher Quality, Parental Involvement, Peer Influence, Parental Education Level, Access to Resources

Binary Variables (Yes/No):
Internet Access, Extracurricular Activities, Learning Disabilities

The target variable for analysis is **Exam Score**, which is used to evaluate academic performance and classify students into grade categories.

Source of Dataset

The dataset has been sourced from a publicly available academic dataset repository (Kaggle). https://www.kaggle.com/datasets/lainguyn123/student-performance-factors/data

It is suitable for exploratory data analysis and performance modeling in the education domain.

C. Project Timeline

A simple table showing planned tasks and expected completion dates.

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding	*Week 1*
2	Data cleaning, preprocessing, and validation / Prompt design	*Week 2*
3	Analysis, AI tool integration (Copilot / Chatbot), and insight generation	*Week 2 – 3*
4	Dashboard / output build, testing, and review	*Week 3*
5	Report writing, GitHub upload, and video recording and submission	*Week 4*

7. Expected Outcomes / Conclusion

By the end of this project, the following outcomes are expected:

• A clean and standardized student performance dataset with documented preprocessing steps.

• A detailed analytical report showing grade distribution and identification of major performance drivers.

• Segmentation of students into performance categories enabling early academic intervention.

• An interactive dashboard providing visual insights into academic trends.

• Demonstrated integration of AI tools for generating summaries and enhancing analytical interpretation.

The project will showcase how structured data analysis combined with AI assistance can significantly improve educational decision-making and student performance monitoring systems.

8. References

- **Microsoft Corporation. Microsoft Copilot. Available at: https://copilot.microsoft.com**

- **Kaggle Inc. Open Datasets. Available at: https://www.kaggle.com/datasets**